

Power Analysis of Subgroup Representation in Clinical Trials: Evidence from the ACTG 320 Trial

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0.1 Abstract

Clinical trials are typically powered to detect treatment effects in the overall study population rather than within demographic subgroups, potentially limiting the ability to identify meaningful differences in treatment effects. Using the ACTG 320 HIV clinical trial as a case study, this paper evaluates how subgroup representation and event counts influence statistical power in time-to-event analyses. Cox proportional hazards models, likelihood ratio tests, Kaplan–Meier curves, and event-driven power analyses were used to evaluate treatment effects overall and across sex and race/ethnicity subgroups. Indinavir-containing therapy significantly reduced the hazard of progression to AIDS or death by approximately 48%, reproducing the original study findings. No statistically significant subgroup interaction effects were detected; however, several demographic subgroups, particularly male, Asian/Pacific Islander, and Other race groups, exhibited unstable hazard ratio estimates due to low event counts and insufficient power. Power simulations demonstrated that substantially larger subgroup sample sizes would be required to reliably detect moderate interaction effects. These findings highlight that statistical power in survival analysis is driven primarily by observed events rather than enrollment counts alone and emphasize the importance of adequate subgroup representation in clinical trial design.

1 Introduction

Randomized clinical trials are typically powered to detect treatment effects in the overall study population rather than within demographic subgroups. This can lead to underrepresentation for female and minority ethnic groups, reducing the effective power of statistical tests and the

ability to detect critical differences in treatment effects. Prior work has shown underrepresentation of women and ethnic minority groups in pivotal clinical trials across therapeutic areas ([Khan et al. 2020](#); [Poon et al. 2013](#); [Eshera et al. 2015](#)).

A key feature of many clinical trials, including ACTG 320, is that outcomes are measured as time-to-event outcomes and analyzed using survival analysis methods. In this framework, statistical inference is driven by the number of observed events (e.g., disease progression or death) instead of the number of participants enrolled. As a result, subgroups with smaller sample sizes tend to yield fewer events, which can substantially reduce the power to detect treatment effects.

This distinction between sample size and event count is central to the present study. Even when treatment effects differ across demographic groups, studies may lack sufficient statistical power to detect these differences if the number of observed events is small. This creates a structural limitation in subgroup analysis that is not always apparent when focusing solely on enrollment counts.

The purpose of this paper is to use the ACTG 320 clinical trial ([Hammer et al. 1997](#)) as a case study to investigate how underrepresentation in gender and ethnic subgroups impacts the statistical power to detect clinically meaningful treatment effects.

The rest of this paper discusses the ACTG320 data set elements and limitations, provides a brief overview of key survival analysis concepts including hazard functions, Cox proportional hazards models and how hypothesis testing and power analysis are applied in this context. The paper concludes with a discussion of the results, implications and areas for further research.

2 Data

The ACTG 320 dataset is derived from a randomized, double-blind clinical trial evaluating the efficacy of combination antiretroviral therapy in HIV-infected patients with advanced disease (CD4 cell counts < 200 cells/mm³). Participants were stratified by their CD4 cell counts at enrollment for those with CD4 cell count < 50 and patients with CD4 cell count 51 - 200 cells/mm³. Patients in each group were randomly assigned to receive either a two-drug nucleoside regimen with a placebo or a three-drug regimen that included the protease inhibitor indinavir.

The study’s primary endpoint is a time-to-event defined as either progression to AIDS or death. A total of 1,156 patients participated in the trial but only 96 experienced progression to AIDS or death during follow-up. Researchers also investigated two secondary endpoints: changes in CD40 cell levels tracked at intervals during the study and changes in plasma HIV-1 RNA concentrations.

The dataset for this paper was obtained from the GLDreg R library (Su et al. 2023). The file contains participant level records that include treatment assignment, baseline clinical characteristics (e.g., CD4 count, prior antiretroviral therapy), demographic variables, and follow-up information on clinical. There are also subgroup indicators (e.g., sex and race/ethnicity), which allow for investigation of treatment effect heterogeneity and, importantly for this study, assessment of how reduced subgroup sample sizes and event counts impact statistical power.

The publicly available data set does not include data on the secondary outcomes so this paper will focus on power analysis on primary endpoint only. Considering that there 96 of enrolled participants experienced the AIDS or death, the power of any subgroup tests is expected to be low, limiting the precision of any subgroup level estimation treatment effects.

A breakdown of the study participants in terms of control vs. treatment groups and number of participants by gender and race/ethnicity is provided.

Table 1: Treatment Group Distribution: Indinavir vs. No Indinavir

Treatment Group	n	%
No Indinavir	577	50.1
Indinavir	574	49.9

Table 2: Subgroup Sample Sizes and Event Rates

Variable	n	Events	%	Event Rate (%)
Sex: Female	951	81	82.6	8.5
Sex: Male	200	15	17.4	7.5
Race: White	596	51	51.8	8.6
Race: Black	327	21	28.4	6.4
Race: Hispanic	203	20	17.6	9.9
Race: Asian/PI	14	3	1.2	21.4
Race: Other	11	1	1.0	9.1

3 Methods

Clinical trials evaluating time-to-event outcomes require statistical methods that properly account for censoring and the timing of events. In the context of the ACTG 320 study, the primary endpoint is the time to progression to AIDS or death, making survival analysis the appropriate framework. The section will start with a review of key concepts in survival analysis and progress to regression model formulations, hypothesis testing and power analysis definitions.

3.1 Survival and Hazard Functions

Let T denote a non-negative random variable representing the time to the event of interest. The survival function is defined as

$$S(t) = P(T > t).$$

This quantity represents the probability that an individual remains event-free beyond time t and provides the basis for survival and hazard analysis. The hazard function is defined as

$$h(t) = \lim_{\Delta t \rightarrow 0} \frac{P(t \leq T < t + \Delta t \mid T > t)}{\Delta t} \quad (1).$$

where Δt is a small time interval, and $h(t)$ represents the instantaneous event rate at time t conditional on survival up to time t (Moore 2016a). The hazard function represents how quickly the survival probability declines over time. Higher values of $h(t)$ correspond to a greater risk of the event occurring immediately after time t , while lower values indicate slower event occurrence. The hazard function can be derived from equation (1) as the ratio of the pdf of T and the survival function

$$h(t) = \frac{f(t)}{S(t)}.$$

3.2 Cox Proportional Hazards Model and Hazard Ratios

The Cox proportional hazards model is used in time-to-event clinical studies to estimate the association between covariates and the event outcomes. The model is semi-parametric since it doesn't require specifying the baseline hazard function. For subject i with covariate vector X_i , the Cox model is

$$h_i(t \mid \mathbf{x}_i) = h_0(t) \exp(\mathbf{x}_i^\top \beta),$$

where $h_0(t)$ is the unspecified baseline hazard function and β is a vector of regression coefficients. In the Cox model, comparisons between groups are made through ratios of hazard functions, allowing treatment effects to be interpreted as relative differences in instantaneous risk over time (Moore 2016a). The hazard ratio (HR) compares the hazard function between two groups

$$HR = \frac{h_T(t)}{h_C(t)}$$

where $h_T(t)$ is the hazard function for the treatment group and $h_C(t)$ hazard function for the control group. An HR less than one indicates the treatment reduces the instantaneous risk and is more effective than the control group while an HR greater than one indicates the converse a detrimental treatment effect. Since the hazard ratios don't depend on t , the Cox model assumes a constant hazard ratio over all time intervals which can be interpreted that the proportion between treatment and control hazard values remains constant.

The Cox proportional hazards model is estimated using partial likelihood estimation to account for censoring of the data (Moore 2016b),

$$L(\beta) = \prod_{i:\delta_i=1} \frac{\exp(X_i^\top \beta)}{\sum_{j \in R(t_i)} \exp(X_j^\top \beta)}.$$

The partial likelihood is converted to the log-likelihood which is maximized to obtain the estimate $\hat{\beta}$.

3.3 Hypothesis Testing for Subgroup Interactions

The likelihood ratio test is generally preferred in survival analysis when evaluating potential changes in interaction effects (Moore 2016b). Let $\ell(\beta)$ denote the log partial likelihood of the Cox model. To test a hypothesis about a subset of parameters, two models are considered:

- A **reduced model**, which imposes the null hypothesis constraints, and
- A **full model**, which includes all parameters of interest.

The likelihood ratio test (LRT) statistic is defined as

$$\Lambda = -2 \left[\ell(\hat{\beta}_{\text{reduced}}) - \ell(\hat{\beta}_{\text{full}}) \right].$$

Assuming the null hypothesis is true, the test statistic follows an approximate chi-square distribution:

$$\Lambda \sim \chi_k^2,$$

where k is the number of restrictions imposed under the null hypothesis. For example, $k = 1$ for a single interaction term and $k > 1$ for joint tests involving multiple subgroup effects. The null hypothesis is rejected at significance level α if

$$\Lambda > \chi_{k, 1-\alpha}^2.$$

3.4 Power Estimation for Event-Based Studies

Power calculations for time-to-event analyses are driven primarily by the number of observed events rather than the total sample size. For a two-group comparison, the approximate number of events required to detect a specified hazard ratio HR with significance level α and power $1 - \beta$ is

For a two-group survival comparison, the approximate power for a fixed number of observed events is

$$\text{Power} = \Phi \left(\sqrt{Dp(1-p)} |\log(HR)| - z_{1-\alpha/2} \right)$$

where D is the observed or planned number of events, p is the proportion assigned to treatment, HR is the target hazard ratio (equivalent to the effect size), α is the two-sided significance level, and $\Phi(\cdot)$ is the standard normal cumulative distribution function ([Schoenfeld 1981](#)).

This expression highlights that the ability to detect treatment effects depends on both the magnitude of the hazard ratio and the total number of events observed during follow-up. Smaller effect sizes require substantially more events to achieve adequate power .

3.5 Primary Treatment Effect Model

The overall treatment effect may be evaluated using a Cox proportional hazards model of the form

$$h_i(t) = h_0(t) \exp \left(\beta_1 T_i + \beta_2^\top Z_i \right)$$

where T_i is the treatment indicator and Z_i is a vector of additional baseline covariates. The coefficient β_1 represents the log hazard ratio for the primary treatment effect after adjusting for the covariates in Z_i . The treatment hazard ratio is

$$HR_T = \exp(\beta_1)$$

The null hypothesis for the overall treatment effect is

$$\begin{aligned} H_0 : \beta_1 = 0 \quad \text{vs.} \quad H_A : \beta_1 \neq 0 \\ \text{or equivalently} \\ H_0 : HR_T = 1 \quad \text{vs.} \quad H_A : HR_T \neq 1 \end{aligned}$$

3.6 Measuring Subgroup Treatment Effects Between Genders

To evaluate whether the treatment effect differs by sex, the regression model is expanded to include a gender indicator and a treatment-by-gender interaction term:

$$h_i(t) = h_0(t) \exp(\beta_1 T_i + \beta_2 G_i + \beta_3 T_i G_i + \beta_4^\top Z_i)$$

where G_i is a binary gender indicator and Z_i represents additional covariates, β_1 is the treatment effect for the reference gender group, β_2 is the baseline difference in hazard between gender groups among participants in the control treatment group, and β_3 is the treatment-by-gender interaction effect.

The hypothesis test for whether the treatment effect differs by gender is

$$\begin{aligned} H_0 : \beta_3 = 0 \quad \text{vs.} \quad H_A : \beta_3 \neq 0 \\ \text{or equivalently} \\ H_0 : \exp(\beta_3) = 1 \quad \text{vs.} \quad H_A : \exp(\beta_3) \neq 1 \end{aligned}$$

3.7 Measuring Subgroup Treatment Effects Between Ethnic Groups

The same interaction framework can be extended to evaluate whether treatment effects differ across ethnic groups. In the ACTG320 trial, ethnicity is coded as: 1 = White non-Hispanic, 2 = Black non-Hispanic, 3 = Hispanic, 4 = Asian or Pacific Islander, 5 = American Indian or Alaska Native, and 6 = Other or unknown.

Using White non-Hispanic participants as the reference group, the model is defined as

$$\begin{aligned} h_i(t) = h_0(t) \exp(\beta_1 T_i + \beta_2 Black_i + \beta_3 Hispanic_i + \beta_4 AsianPI_i + \beta_5 AIAN_i + \beta_6 Other_i \\ + \beta_7 T_i Black_i + \beta_8 T_i Hispanic_i + \beta_9 T_i AsianPI_i + \beta_{10} T_i AIAN_i + \beta_{11} T_i Other_i + \beta_{12}^\top Z_i) \end{aligned}$$

In this model, β_1 is the treatment effect for White non-Hispanic participants. The coefficients β_2 through β_6 represent baseline hazard differences between each ethnic subgroup and the White non-Hispanic reference group among participants in the control treatment group.

The interaction coefficients β_7 through β_{11} measure whether the treatment effect differs between each ethnic subgroup and the White non-Hispanic reference group. The vector Z_i includes additional baseline covariates.

The joint hypothesis test for whether treatment effects differ across ethnic groups is

$$H_0 : \beta_7 = \beta_8 = \beta_9 = \beta_{10} = \beta_{11} = 0$$

$$H_A : \text{At least one interaction coefficient is not equal to zero}$$

If the null hypothesis is rejected, there is evidence that the treatment effect differs for at least one ethnic subgroup relative to the White non-Hispanic reference group.

3.8 Power Analysis for Treatment and Subgroup Effects

Since our effect is based on hazard functions, power calculations for survival outcomes are based on the number of observed events instead of sample size. Under the Cox proportional hazards model, the statistical power to detect treatment effects is driven by the magnitude of the hazard ratio and the total number of events observed.

3.9 Power for the Overall Treatment Effect

For a two-group survival comparison, power can be estimated directly when the number of observed or anticipated events is known. This formulation is useful for sensitivity analysis because the available event count may be fixed by the completed study or by the expected duration of follow-up.

For a two-sided test at significance level α , the approximate power is

$$\text{Power} = \Phi \left(\sqrt{Dp(1-p)} |\log(HR)| - z_{1-\alpha/2} \right),$$

where D is the total number of observed or anticipated events, p is the proportion assigned to the treatment group, $1-p$ is the proportion assigned to the control group, HR is the target hazard ratio comparing treatment with control, $z_{1-\alpha/2}$ is the standard normal critical value for a two-sided test, and $\Phi(\cdot)$ is the standard normal cumulative distribution function ([Schoenfeld 1981](#)).

3.10 Power for Subgroup Differences: Gender

Since interaction effects are estimated from contrasts between subgroup-specific treatment effects, the effective amount of information is reduced relative to the overall treatment comparison. Consequently, substantially larger event counts are typically required to detect treatment effect differences across subgroups ([Schoenfeld 1981](#)).

For a two-sided interaction test at significance level α , the approximate power is

$$\text{Power} = \Phi \left(\sqrt{D p_T(1-p_T) p_G(1-p_G)} |\log(HR_{\text{int}})| - z_{1-\alpha/2} \right),$$

where D is the total number of observed or anticipated events, p_T is the proportion assigned to the treatment group, p_G is the proportion of participants in one of the subgroup categories (for example, one gender subgroup), HR_{int} is the interaction hazard ratio, $z_{1-\alpha/2}$ is the standard normal critical value for a two-sided test, and $\Phi(\cdot)$ is the standard normal cumulative distribution function (Schoenfeld 1981).

The interaction hazard ratio quantifies the relative difference between subgroup-specific treatment effects. For example,

$$HR_{\text{int}} = \frac{HR_1}{HR_0}.$$

3.11 Subgroup Power Analysis: Ethnicity

For a two-sided interaction test at significance level α , the approximate power is

$$\text{Power} = \Phi \left(\sqrt{D p_T (1 - p_T) p_E (1 - p_E)} |\log(HR_{\text{int}})| - z_{1-\alpha/2} \right),$$

p_T is the proportion assigned to the treatment group, p_E is the proportion of participants in one of the ethnic subgroup categories, HR_{int} is the interaction hazard ratio comparing subgroup-specific treatment effects, $z_{1-\alpha/2}$ is the standard normal critical value for a two-sided test, and $\Phi(\cdot)$ is the standard normal cumulative distribution function (Schoenfeld 1981).

4 Results

The analysis is organized into four sections: whole group, sex subgroup, race/ethnicity subgroup and power analysis. Analysis and exhibits in this paper was generated using the R statistical package (R Core Team 2026) primarily using the survival library (Therneau 2026).

4.1 Whole Group

The Kaplan-Meier curves for the control and treatment groups are shown and visually suggest a noticeable reduction in the hazard.

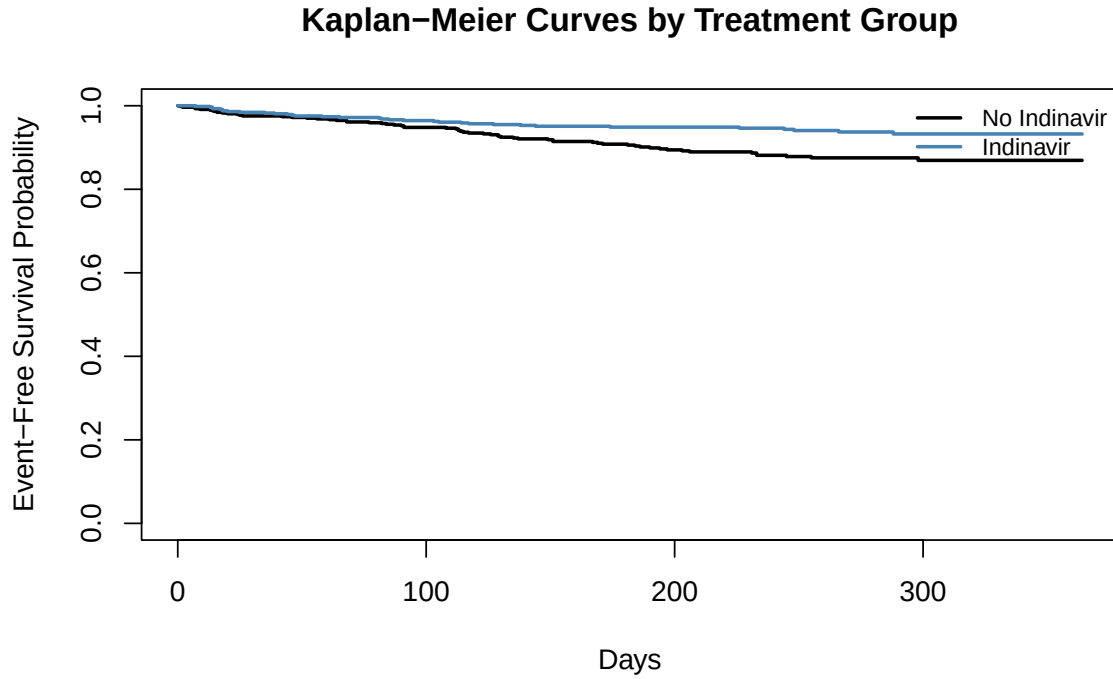


Figure 1: Kaplan-Meier survival curves for Indinavir versus No Indinavir treatment groups.

The baseline Cox model includes the following covariates: treatment indicator (tx), baseline CD4 count (cd4), age, Karonfsky performance scale (karnof), months of prior ZDV use (priorzdv) and CD40 cell count stratification (strat2). The regression results are shown in the following table.

An $HR = 0.517$ for the treatmen indicator means that at any given point in time, patients receiving indinavir had about half the hazard of experiencing an AIDS-defining event or death compared to patients not receiving indinavir after controlling for the other covariates, which is consistent with what Hammer et al. (1997) reported and confirms your model is reproducing the original study findings.

Table 3: Whole-Group Baseline Cox Model — Hazard Ratios

Covariate	HR	95% CI		p-value
		Lower	Upper	
Treatment: Indinavir vs. No Indinavir	0.517	0.339	0.788	0.002
Age	1.022	1.000	1.045	0.054
Baseline CD4	0.985	0.978	0.993	0.000

Covariate	HR	95% CI Lower	95% CI Upper	p-value
Karnofsky Score	0.948	0.926	0.970	0.000
Prior ZDV Use	1.000	0.992	1.007	0.925
CD4 Stratum	1.012	0.517	1.984	0.971

A log-rank test was used to test overall model significance and based on the chi-square statistic of 10.545 with 1 degree of freedom yields $p = 0.0012$, which is highly statistically significant indicating that patients receiving indinavir had meaningfully different event timing than those who did not.

4.2 Gender Subgroup

To test if there is a difference in the treatment effect for men vs. women in the group, two models are fit. The baseline model adds participant `sex` as a fixed covariate alongside treatment. The full model adds a `tx * sex` interaction term to test whether sex *modifies* the treatment effect. A likelihood ratio test compares the two models and then two Cox models are fit within each sex stratum (female-only, male-only) to calculate stratum-specific HRs.

4.2.1 Gender Baseline Model

Sex is included as a covariate with no interaction, estimating the marginal treatment effect after adjusting for sex. Results shown below show the HR remains unchanged at 0.517. The HR for sex has a highly insignificant p-value indicating that sex has no meaningful impact on the hazard ratio.

Table 4: Sex Baseline Cox Model — Hazard Ratios

Covariate	HR	95% CI Lower	95% CI Upper	p-value
Treatment (IDV vs no IDV)	0.517	0.339	0.789	0.002
Sex (Male vs Female)	1.122	0.643	1.957	0.685
Age	1.023	1.000	1.046	0.050
Baseline CD4	0.985	0.978	0.993	0.000
Karnofsky Score	0.948	0.926	0.970	0.000
Prior ZDV Use	1.000	0.992	1.007	0.926
CD4 Stratum	1.013	0.517	1.985	0.970

4.2.2 Sex Full Model (Interaction)

In the full model, the HR for treatment among females (the reference group for sex) is 0.460. This means females receiving indinavir have about 54% lower hazard than females not receiving it which is a statistically significant difference. The interaction HR of 2.034 means the treatment effect in males which means protective effect of indinavir appears *weaker* in males than females. However the 95% CI is very wide and crosses 1.0, and $p = 0.2141$, so this difference is **not statistically significant**. You cannot conclude from this data that sex genuinely modifies the treatment effect.

Table 5: Sex Full Cox Model with Interaction — Hazard Ratios

Covariate	HR	95% CI Lower	95% CI Upper	p-value
Treatment (IDV vs no IDV)	0.460	0.288	0.734	0.001
Sex (Male vs Female)	0.854	0.405	1.801	0.679
Age	1.023	1.000	1.046	0.050
Baseline CD4	0.985	0.978	0.993	0.000
Karnofsky Score	0.947	0.925	0.969	0.000
Prior ZDV Use	0.999	0.992	1.007	0.865
CD4 Stratum	1.011	0.516	1.978	0.975
Treatment $\times$ Sex	2.034	0.663	6.238	0.214

The likelihood ratio test confirms this with a $p\text{-value} = 0.2187$. Adding the interaction term does not significantly improve model fit over the baseline model so we fail to reject the null hypothesis that the treatment effect is meaningfully different between female and male patients. The effect sizes by set are estimated and shown below.

With only 15 events in the male stratum, the Cox model has very little information to work with. A general rule of thumb in survival analysis is that you need at least 10–15 events per predictor variable in the model. Your male stratum has 15 events total across 6 predictors — the model is severely underpowered and the estimates are unreliable. For comparison, the Kaplan-Meier survival curves are shown below.

4.3 Race/Ethnicity Subgroup

Following the same procedure as with sex subgroups, a baseline with no interactions and full model with interactions are developed and tested. In this case, race/ethnicity is included as a 5-level factor with White as the reference group.

4.3.1 Baseline Race/Ethnicity Model

The treatment HR = 0.526 means that after adjusting for race/ethnicity as a 5-level factor, patients receiving indinavir had approximately 47% lower hazard of AIDS-defining events or death compared to patients not receiving indinavir. This result suggests that race/ethnicity is a not meaningful confounder of the treatment-outcome relationship like the participant's gender.

4.3.2 Race Full Model

The treatment HR for White patients (the reference group) is 0.534, confirming a significant protective effect in White patients specifically.

Among all of the race interaction terms, Black, Hispanic and Other are all insignificant, suggesting no meaningful difference in treatment effect by race. At first glance, Asian/PI appears to be the exception appearing to be highly significant but looking at the bounds of the 95% CI suggests highly unstable estimates due extremely small sample size.

This is confirmed with the results of the likelihood ratio test with Chi-square = 8.326, df = 4, p = 0.080. The interaction as a whole is **not statistically significant** at the conventional $\alpha = 0.05$ threshold, though it approaches significance at $\alpha = 0.10$. Formally, it cannot be concluded that race/ethnicity modifies the treatment effect. The borderline p-value is largely being driven by the unstable Asian/PI interaction term. Excluding these observations from the data would reduce the significance level.

The race/ethnicity subgroup analysis is limited by severe sample size constraints in the non-White groups, particularly Asian/PI and Other. The formally correct conclusion is that race/ethnicity does not significantly modify the treatment effect at $\alpha = 0.05$.

Table 6: Race Full Cox Model with Interaction — Hazard Ratios

Covariate	HR	95% CI Lower	95% CI Upper	p-value
Treatment (IDV vs no IDV)	0.534	0.302	0.946	0.032
Race: Black (vs White)	0.794	0.432	1.457	0.456
Race: Hispanic (vs White)	1.002	0.524	1.914	0.995
Race: Asian/PI (vs White)	0.860	0.117	6.317	0.882
Race: Other (vs White)	2.831	0.378	21.227	0.311
Age	1.024	1.001	1.048	0.038
Baseline CD4	0.985	0.978	0.993	0.000
Karnofsky Score	0.948	0.927	0.971	0.000
Prior ZDV Use	0.997	0.989	1.005	0.467
CD4 Stratum	1.010	0.513	1.986	0.978

Covariate	HR	95% CI Lower	95% CI Upper	p-value
Treatment $\times$ Black	0.591	0.185	1.883	0.373
Treatment $\times$ Hispanic	1.249	0.422	3.698	0.687
Treatment $\times$ Asian/PI	15.514	1.244	193.477	0.033
Treatment $\times$ Other	0.000	0.000	Inf	0.993

Below are the estimated race specific effect sizes.

4.4 Summary of All Effect Sizes

The following table summarizes all effect sizes as measured by HR. As can be seen they are stable for overall group, sex and race but become unreliable at the subgroup level due to the small number of events and in the case of Asian/PI and Other the estimates are completely unreliable.

HR < 1 indicates indinavir-containing regimens reduce the hazard of AIDS-defining events or death relative to nucleoside-only therapy. Interaction p-values from the LR tests indicate whether subgroup membership significantly modifies the treatment effect. Note that the Asian/PI and Other race subgroups have small sample sizes; confidence intervals for these groups should be interpreted with caution.

Table 7: Summary of Treatment Effect Sizes Across All Analyses

Analysis	Model	Subgroup	n	Events	HR	95% CI Lower	95% CI Upper
Whole Group	Baseline	All	1151	96	0.517	0.339	0.788
Sex	Baseline (pooled)	All	1151	96	0.517	0.339	0.789
Sex	Stratum-specific	Female	951	81	0.459	0.288	0.732
Sex	Stratum-specific	Male	200	15	1.101	0.393	3.087
Race	Baseline (pooled)	All	1151	96	0.526	0.344	0.805
Race	Stratum-specific	White	596	51	0.539	0.303	0.960
Race	Stratum-specific	Black	327	21	0.287	0.104	0.794
Race	Stratum-specific	Hispanic	203	20	0.628	0.246	1.602

Analysis	Model	Subgroup	n	Events	HR	95% CI Lower	95% CI Upper
Race	Stratum-specific	Asian/PI	14	3	0.068	0.000	Inf
Race	Stratum-specific	Other	11	1	237.311	0.000	Inf

4.5 Initial Observed Power for ACTG 320 Subgroup Analyses

Before estimating the sample sizes needed for future studies, the observed or **effective power** of the ACTG 320 analyses was calculated for the overall study population and for each sex and race/ethnicity subgroup.

Power was estimated assuming $p = 0.50$ is the assumed treatment allocation proportion $\alpha = 0.025$ was used to account for the Bonferroni-adjusted two-sided significance level. These calculations should be interpreted descriptively because observed post hoc power is a function of the estimated effect size and event count.

Table 8: Initial Observed Power by Overall Group and Subgroup

Analysis	Subgroup	n	Events	Observed HR	Event Rate	Observed Power	Interpretation
Whole Group	All	1151	96	0.517	0.083	0.839	At or above 80%
Sex	Female	951	81	0.459	0.085	0.897	At or above 80%
Sex	Male	200	15	1.101	0.075	0.020	Below 80%
Race/Ethnicity	White	596	51	0.539	0.086	0.486	Below 80%
Race/Ethnicity	Black	327	21	0.287	0.064	0.732	Below 80%
Race/Ethnicity	Hispanic	203	20	0.628	0.099	0.115	Below 80%
Race/Ethnicity	Asian/PI	14	3	0.068	0.214	0.535	Below 80%
Race/Ethnicity	Other	11	1	237.311	0.091	0.689	Below 80%

The initial power estimates show that the overall study and some larger subgroups had adequate power to detect the observed treatment effect, while smaller subgroups were limited by sparse event counts. In particular, estimates for male, Asian/Pacific Islander, and Other race groups should be interpreted cautiously because their hazard ratios were estimated from very few events, making the calculated power unstable and primarily descriptive.

4.6 Power Analysis: Required Sample Sizes for a Future Study

The subgroup analyses in revealed substantial imbalance in sample sizes across sex and race/ethnicity groups, leaving several subgroups severely underpowered. This section uses the observed effect sizes (HRs) from the stratum-specific models as assumed true effects in a prospective power analysis, estimating the sample sizes a future study would need to reliably detect those effects.

For subgroups with unstable HR estimates (Asian/PI, Other, Male) the analysis uses a **range of plausible HRs** rather than the unreliable observed estimate, to illustrate what sample sizes would be needed to detect effects of various magnitudes.

4.6.1 Reliable Subgroups

These subgroups had sufficient events for stable HR estimation.

Table 9: Required Sample Sizes by Subgroup and Power Level ($\alpha = 0.025$)

Subgroup	Observed HR	Obs. n	Obs. Events	Event Rate	n (80%)	n (85%)	n (90%)	n (95%)
Whole Group	0.517	1151	96	0.083	264	300	348	420
Sex: Female	0.459	951	81	0.085	188	212	247	294
Race: White	0.539	596	51	0.086	293	339	386	468
Race: Black	0.287	327	21	0.064	109	109	125	156
Race: Hispanic	0.628	203	20	0.099	447	508	589	711

4.6.2 Underpowered Subgroups — Plausible HR Range

For the male, Asian/PI, and Other subgroups, the observed HRs were unreliable due to very small event counts. Instead, this table shows required sample sizes across a **range of plausible HRs** (0.50 to 0.90), using each subgroup's observed event rate. This illustrates what a future study would need to enroll to detect effects of various magnitudes in these groups.

Table 10: Required Sample Sizes for 80% Power Across Assumed Hazard Ratios

Subgroup	Assumed HR	Observed Event Rate	Required n for 80% Power
Sex: Male	0.5	0.075	267
Sex: Male	0.6	0.075	494

Subgroup	Assumed HR	Observed Event Rate	Required n for 80% Power
Sex: Male	0.7	0.075	1000
Sex: Male	0.8	0.075	2547
Sex: Male	0.9	0.075	11427
Race: Asian/PI	0.5	0.214	94
Race: Asian/PI	0.6	0.214	173
Race: Asian/PI	0.7	0.214	350
Race: Asian/PI	0.8	0.214	892
Race: Asian/PI	0.9	0.214	4000
Race: Other	0.5	0.091	220
Race: Other	0.6	0.091	407
Race: Other	0.7	0.091	825
Race: Other	0.8	0.091	2101
Race: Other	0.9	0.091	9427

4.6.3 Events Required vs. Sample Size

This table separates the required **number of events** from the required **total sample size** for the reliable subgroups at 80% power, to illustrate how event rate affects study planning.

Table 11: Required Events and Sample Sizes for 80% Power

Subgroup	Observed HR	Event Rate	Events Required for 80% Power	Required n for 80% Power
Whole Group	0.517	0.083	22	264
Sex: Female	0.459	0.085	16	188
Race: White	0.539	0.086	25	293
Race: Black	0.287	0.064	7	109
Race: Hispanic	0.628	0.099	44	447

4.6.4 Power Curve Plot

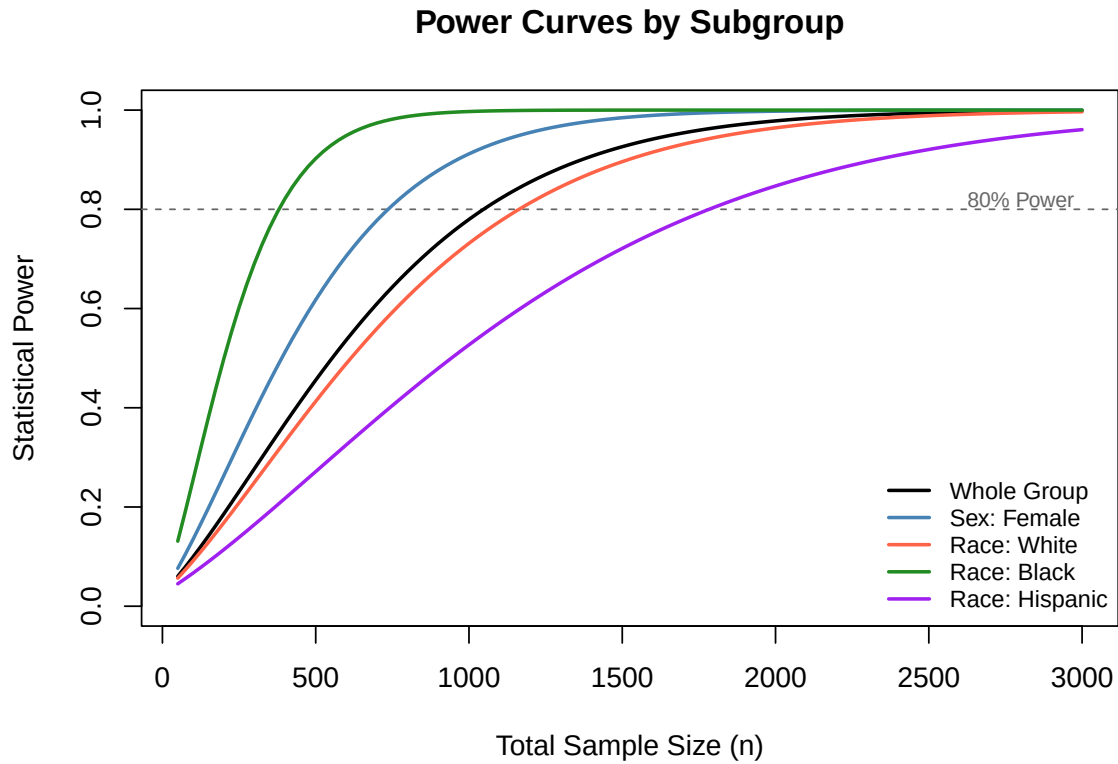


Figure 2: Power curves showing achieved power as a function of sample size for each reliable subgroup, with the 80% power threshold shown.

4.6.5 Sample Sizes Required for 80% Power

Table 12: Adequacy of ACTG 320 Sample Sizes for 80% Power

Subgroup	Observed HR	ACTG 320 n	Required n for 80% Power	ACTG 320 Status
Whole Group	0.517	1151	264	Adequate
Sex: Female	0.459	951	188	Adequate
Race: White	0.539	596	293	Adequate
Race: Black	0.287	327	109	Adequate
Race: Hispanic	0.628	203	447	Underpowered

5 Conclusion

This study evaluated how subgroup representation influences the statistical power to detect treatment effect differences in time-to-event clinical trials using the ACTG 320 HIV trial as a case study. The analysis reproduced the primary findings of the original study, demonstrating that the addition of indinavir substantially reduced the hazard of progression to AIDS or death relative to the control group therapy across all baseline models.

The overall treatment effect was estimated with reasonable precision and power, subgroup analyses revealed substantial limitations associated with reduced sample size and low event counts. The female and White subgroups contained sufficient numbers of events to produce stable hazard ratio estimates and narrow confidence intervals. In contrast, the male, Asian/Pacific Islander, and Other race subgroups experienced very small numbers of observed events, leading to highly unstable effect estimates.

Overall, the treatment effects for sex and race were insignificant and failed to support any assertion for statistically significant differences in subgroup treatment effects. This was further supported in the power analysis by demonstrating that adequate subgroup inference in survival analysis depends primarily on the number of observed events rather than the total enrollment size alone. Subgroups with moderate event rates but small enrollment sizes required substantially larger future sample sizes to achieve conventional levels of statistical power. The Hispanic subgroup, for example, produced stable but underpowered results despite having more than 200 enrolled participants because relatively few events occurred during follow-up. Analysis showed that some underrepresented subgroups would require several-fold increases in enrollment to achieve (80%) power for subgroup-specific analyses.

These findings have broader implications for the design and interpretation of clinical trials. Many pivotal clinical trials are appropriately powered for overall treatment comparisons but are not designed to support reliable subgroup inference. As a result, conclusions regarding subgroup differences are often limited by insufficient information rather than the absence of biological or clinical differences. This issue is particularly important in studies involving demographic groups that have historically experienced lower representation in clinical research.

Several limitations should be acknowledged. First, the analysis relied on secondary use of publicly available ACTG 320 data and therefore was limited to variables available within the released dataset. Second, the total number of primary endpoint events was relatively small, restricting the precision of subgroup analyses. Third, the subgroup power calculations relied on asymptotic approximations based on Cox proportional hazards models and assumed proportional hazards over time. Finally, several subgroup estimates were unstable due to sparse-event conditions, limiting interpretability.

Future research should investigate trial-design strategies that improve subgroup power while maintaining feasible enrollment targets. Potential approaches include oversampling underrepresented populations, extending follow-up duration to increase event accrual, adaptive event-driven designs, Bayesian hierarchical modeling, and pooled meta-analytic subgroup frameworks

across multiple studies. Additional work may also evaluate how alternative survival modeling approaches perform under sparse-event subgroup conditions.

Overall, this study demonstrates that subgroup underrepresentation in clinical trials can substantially impair the ability to detect meaningful treatment effect heterogeneity in survival analyses. Although ACTG 320 was sufficiently powered to identify the overall therapeutic benefit of indinavir, several demographic subgroup analyses remained underpowered due to limited event counts. These findings reinforce the importance of considering subgroup representation and anticipated event accrual during clinical trial planning when subgroup-specific inference is a primary scientific objective.

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